

VINODH BALAKRISHNA

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PROFESSIONAL SUMMARY

Embedded Systems Software Engineer with deep expertise in low-level programming, OS internals, and Linux-based platforms. Experienced in firmware development, hardware bring-up, and debugging complex system-level issues across ARM and x86 architectures. Skilled at writing clean, efficient code and bridging the hardware-software boundary. A focused problem-solver with a curious mindset, thriving both independently and in collaborative, fast-paced environments to deliver reliable, maintainable solutions.

EXPERIENCE

Embedded Linux Engineer

Chennai, India

Comcast Corporation

Dec 2021 - Jul 2023

- Contributed to the Platform Security team by leveraging the Yocto Project build system and RDK software stack to create and manage custom Linux firmware for over 1 million CPE devices, including set-top boxes, routers, and TVs.
- Secured and optimized a custom Restricted Shell with a multi-threaded architecture by implementing command logging, unauthorized command blocking, and tab completion, boosting efficiency and usability by 40%.
- Migrated SBOM functionality from Yocto Kirkstone to Dunfell to enhance supply chain security and maintain 100% software inventory transparency.
- Renewed and rotated 5+ X.509 PKI certificates, ensuring uninterrupted secure TLS communication with zero certificate expiry incidents during release cycles.
- Performed key platform security enhancements to RDKB devices by enabling ASLR with position-independent code and securely removing video analytics module, improving firmware performance by 25% while reducing maintenance overhead.
- Accessed Linux-based CPE devices on ARM and Atom platforms for board/device bring-up and firmware debugging by connecting via Raspberry Pi serial console using Minicom with appropriate baud rate settings.
- Used CI/CD pipelines with Jenkins and Shell scripting to trigger and monitor automated builds and tests in an Agile environment, while utilizing Gerrit for code commits and reviews.

TECHNICAL PROJECTS

Torrent CLI - BitTorrent Client | Source Code: https://github.com/v1n0dh/torrent_cli

- Developed a high-performance BitTorrent client in modern C++23, enabling efficient concurrent downloads and full support for the BitTorrent protocol including peer communication, tracker requests, and piece/block management.
- Implemented zero-copy memory-mapped I/O using `mmap` and `RAII`, thread-safe message passing with custom `Shared_Queue` classes, and concurrency control using mutexes and futures to coordinate peer threads.
- Leveraged ASIO library for scalable asynchronous networking to handle multiple peer connections with fine-grained control, and designed full Bencode encoder/decoder for .torrent metadata serialization and deserialization.
- Used debugging and analysis tools including Wireshark for network traffic inspection, Valgrind for memory diagnostics, and strace to trace syscalls and debug low-level I/O behavior.

Custom Linux Shell (Vshell) | Source Code: <https://github.com/v1n0dh/vshell>

- Built a custom shell in C++, inspired by a coding challenge at codingchallenges.fyi, closely resembling bash, completing the project within 2 weeks to demonstrate core shell functionalities.
- Incorporated key shell features, including command execution, I/O redirection, and pipes, facilitating efficient user interaction with the system.
- Integrated tab completion for files and command history, enhancing user experience and productivity.

Real-Time System Metrics Statusbar | Source Code: <https://github.com/v1n0dh/dwmstatus>

- Created a custom status bar for the dwm window manager in C that displays real-time system metrics such as network speeds, Wi-Fi SSID, CPU temperature, RAM usage, battery status, and current time.
- Collected system data by directly reading low-level kernel interfaces under `/sys/class` for power supply, thermal zones, memory, and network statistics, and used X11 library to dynamically update the status bar on the dwm.
- Implemented network speed calculation by tracking changes in RX/TX bytes per second from network interface statistics, and used wireless extensions `ioctl` calls to fetch Wi-Fi SSID, providing live connectivity information.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Shell Scripting, Java, Rust, x86 Assembly

Core Competencies: Data Structures & Algorithms, Memory Management, Concurrency (Multi-threading/Processing), IPC

Operating Systems & Embedded: GNU Linux/Unix Systems, Embedded Linux Firmware, RTOS

Networking & Communication: TCP/IP, UDP, SNMP, HTTP, TLS/SSL, Networking Protocols, UART, SPI, I2C

Frameworks & Libraries: Yocto Project, RDK, GNU Readline, Jsoncpp, ASIO (non-Boost)

Tools: CMake, GDB, QEMU, OpenSSL, Wireshark, Jenkins, Git, Splunk, Vim, Emacs

EDUCATION

University of Central Missouri

Master's in Computer Science

Coursework: Operating Systems, Algorithms, Computer Networking & Security, Compiler Design & Construction, Cryptographic Algorithms and Protocols, Web Application Security, Cloud Security

Bharath Institute of Higher Education and Research

B.Tech in Computer Science Engineering

Lee's Summit, MO

Aug 2023 - May 2025

Chennai, India

Jul 2017 - May 2021

CERTIFICATIONS

- Rust Essential Training by Barron Stone, LinkedIn Learning - [Certificate Link](#)
- Linux Device Drivers by Kevin Dankwardt, LinkedIn Learning - [Certificate Link](#)